

Programming for Problem Solving

By Boina Avinash Kumar (M.Tech CSI)
Assistant Professor

Outline

- * Algorithm
- * Properties
- * Examples
- * Flowchart
- * Examples



How To Prepare?

2 Minutes to Great Taste

1



Break the Noodle cake into four parts.

2



Add Noodles and "TASTEMAKER" to 210 ml (approx. 1½ tea cups) of boiling water.

3



Cook for 2 minutes in an open pan. Stir occasionally. Do not drain remaining water. Hold for 1 min after cooking. Serve hot.

STORAGE INSTRUCTIONS



Store the remaining noodle cakes in an air tight container to retain freshness.



Algorithm

- * **Step by step procedure for solving any problem** is known as algorithm.
- * An algorithm is a finite set of instructions that, if followed, **accomplishes a particular task.**
- * An algorithm is a sequence of **computational steps** that **transform the input into a valuable** or required **output.**

Properties

1. **Input:** There are more **quantities** that are extremely **supplied**.
2. **Output:** At least **one quantity** is **produced**.
3. **Definiteness:** Each instruction of the algorithm should **be clear and unambiguous**.
4. **Finiteness:** The **process** should be **terminated** after a **finite number of steps**.
5. **Effectiveness:** Every instruction must be **basic enough to be carried out theoretically** or by using paper and pencil

Examples

Algorithm for Addition of two numbers

Step 1: Start

Step 2: Declare $sum=0$

Step 3: Read values for num1, num2

Step 4: Add num1 and num2 and assign the result to sum

Step 5: Display sum

Step 6: Stop

Examples

Algorithm to convert temperature from Celsius(°C) to Fahrenheit(°F)

Step 1: Start

Step 2: Declare $F=0$

Step 3: Read the temperature value in Celsius

Step 3: **Applying Formula of Conversion i.e, $F=(9*C)/5+32$**

Step 4: Print the Temperature in Fahrenheit

Step 5: Stop

Examples

Algorithm to find maximum of two numbers

Step 1: Start

Step 2: Read num1 and num2

Step 3: Compare num1 with num2

If num1 is greater than num2, num1 is greatest
else num2 is greater

Step 4: Print maximum number

Step 5: Stop

Examples

Algorithm to find even number

Step 1: Start

Step 2: Read a number n

Step 3: Check the number n is even or odd

If $n \% 2$ is equal to 0, then n is even

else n is odd number.

Step 4: Print a message that n is even number

Step 5: Stop

Examples

Algorithm to find sum of n numbers

Step 1: Start

Step 2: Declare $\text{sum}=0, i=0$

Step 3: Read number num

Step 4: Repeat the step 5 to step 6 until $i \leq \text{num}$

Step 5: $\text{sum}=\text{sum}+i$

Step 6: increment i

Step 7: Print sum

Step 8: Stop

Question

Write an algorithm for finding **Area of a circle**

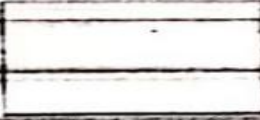
Write an algorithm **to convert temperature from Fahrenheit($^{\circ}\text{F}$) to Celsius($^{\circ}\text{C}$)**

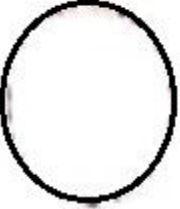
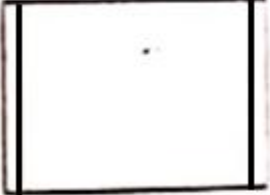
Write an algorithm to find **distance between two points**

Write an algorithm to find **perimeter and area of a rectangle**

Flowchart

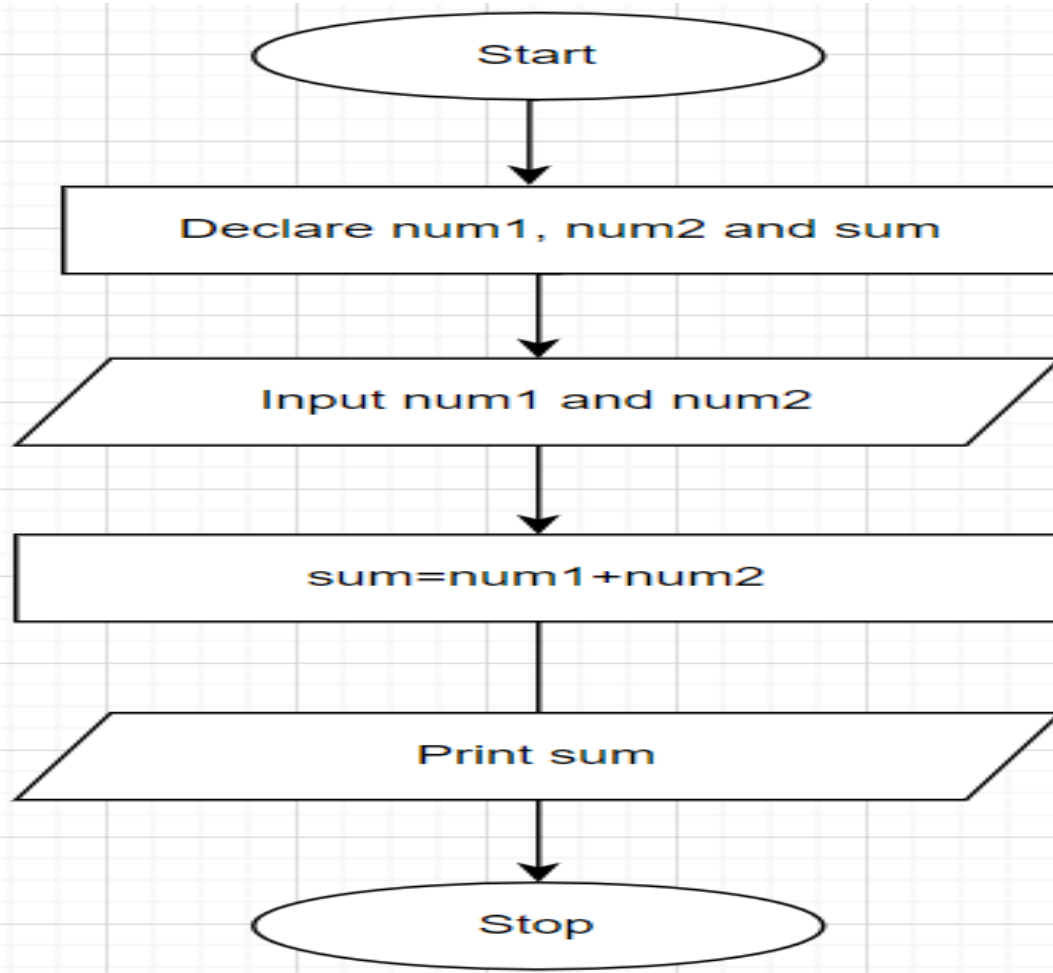
- * A flowchart is a **graphical representation** of an **algorithm**.
- * Programmers often use it as a **program-planning tool** to **solve a problem**.
- * It makes use of **symbols** that are connected among them to indicate the flow of information and processing.
- * The process of drawing a flowchart for an algorithm is known as “**flowcharting**”.

Symbol	Purpose	Description
	Flow line	Used to indicate the flow of logic by connecting symbols.
	Terminal(Stop/Start)	Used to represent start and end of flowchart.
	Input/output	Used for input and output operation.
	Processing	Used for arithmetic operations and data-manipulations.
	Decision	Used to represent the operation in which there are two alternatives, true and false.

Symbol	Purpose	Description
	On-page Connector	Used to join different flow line
	Off-page Connector	Used to connect flowchart portion on different page.
	Predefined Process/Function	Used to represent a group of statements performing one processing task.

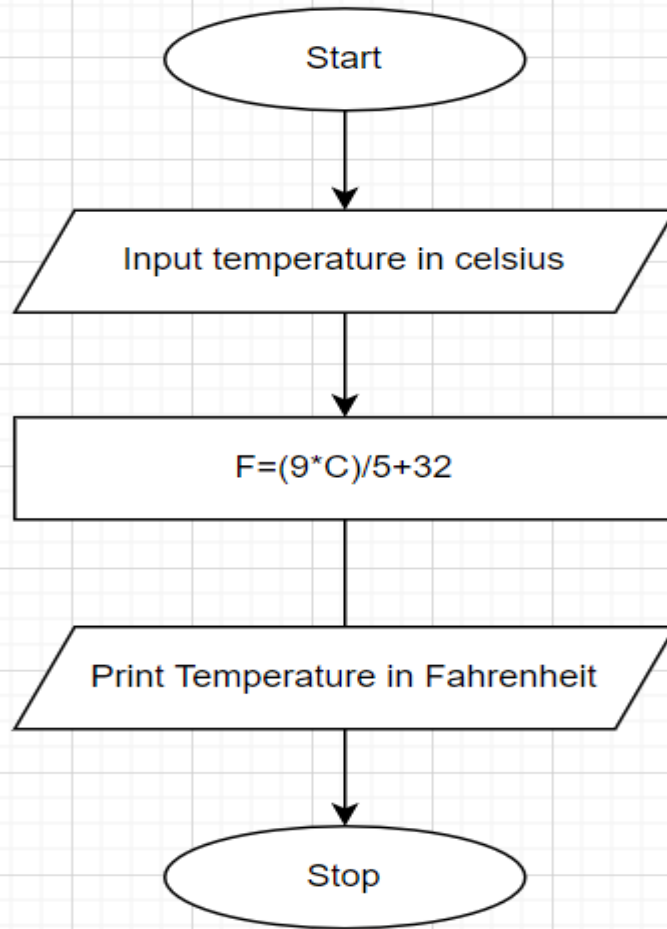


Flowchart for Addition of two numbers



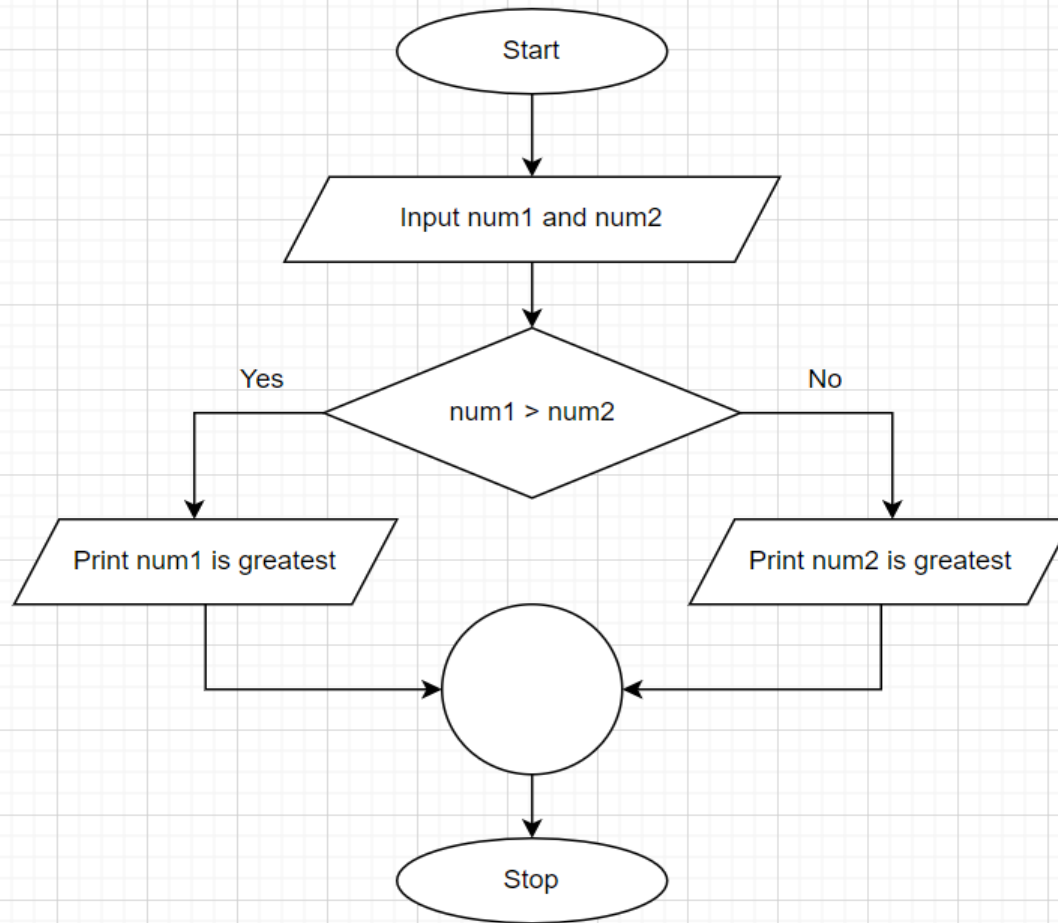


Flowchart to convert temperature from Celsius($^{\circ}\text{C}$) to Fahrenheit($^{\circ}\text{F}$)



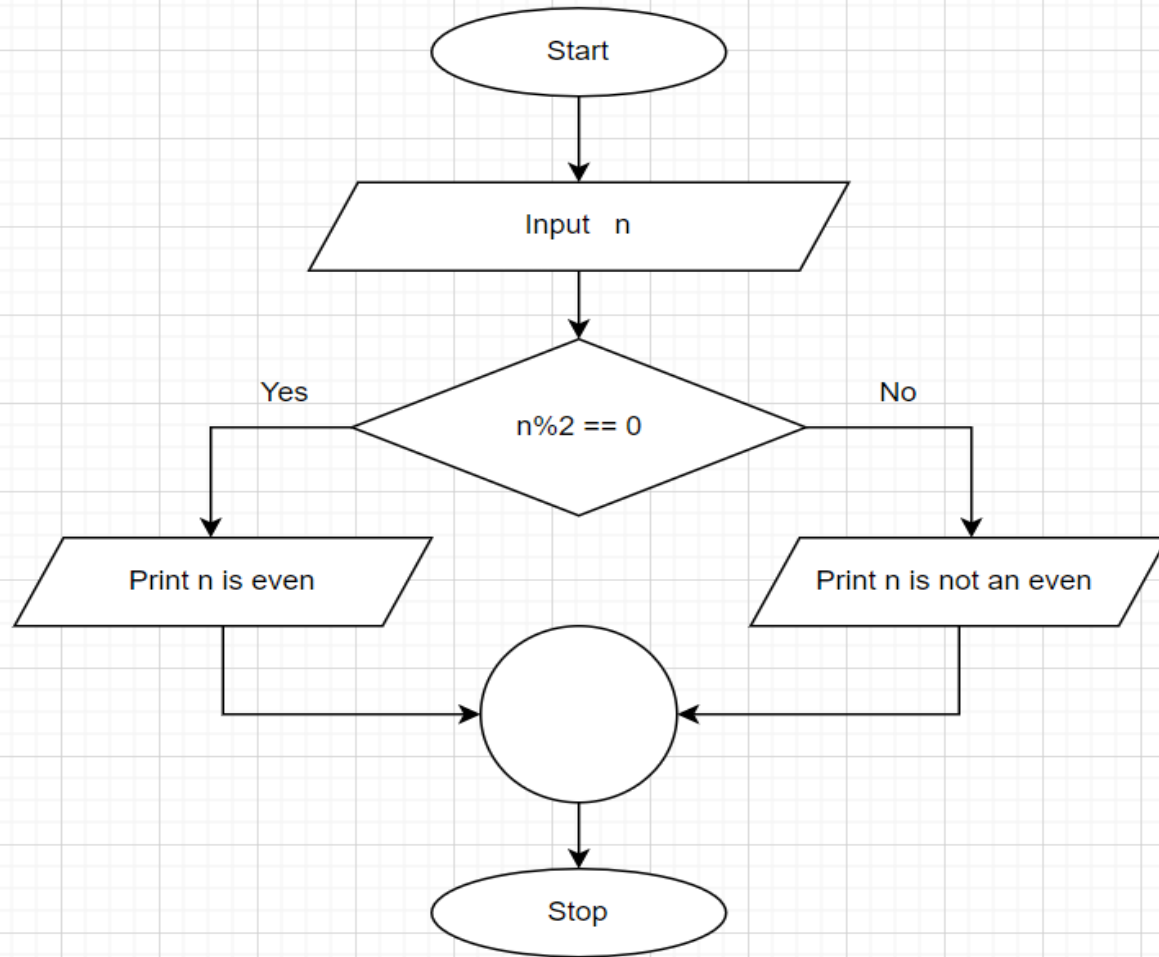


Flowchart to input two numbers from the user and display the largest of two numbers





Flowchart to find even number





Thank You